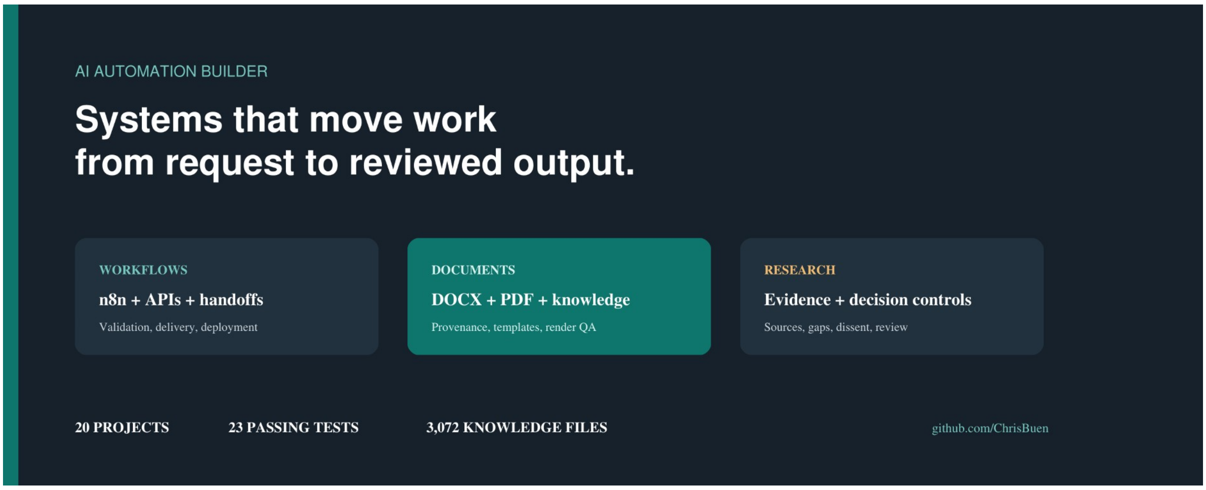
# ChrisBuen

## *AI Automation Builder — evidence-led casebook*

I design AI-assisted systems that move work from an unstructured request to a controlled output: a retrieved file, a research finding, a review-ready document, a decision brief, a chart, or a content package.

This casebook is organized around proof. The three flagship projects include sanitized code, workflow structure, fictional fixtures, recorded implementation results, and runnable public tests. The remaining work shows breadth without giving every project the same evidence label.




Portfolio overview

**Public contact:** `github.com/ChrisBuen` · **Portfolio scope:** 20 project folders · 23 passing public tests · editable sources plus PDF/DOCX outputs

# The portfolio in one view

## What I build

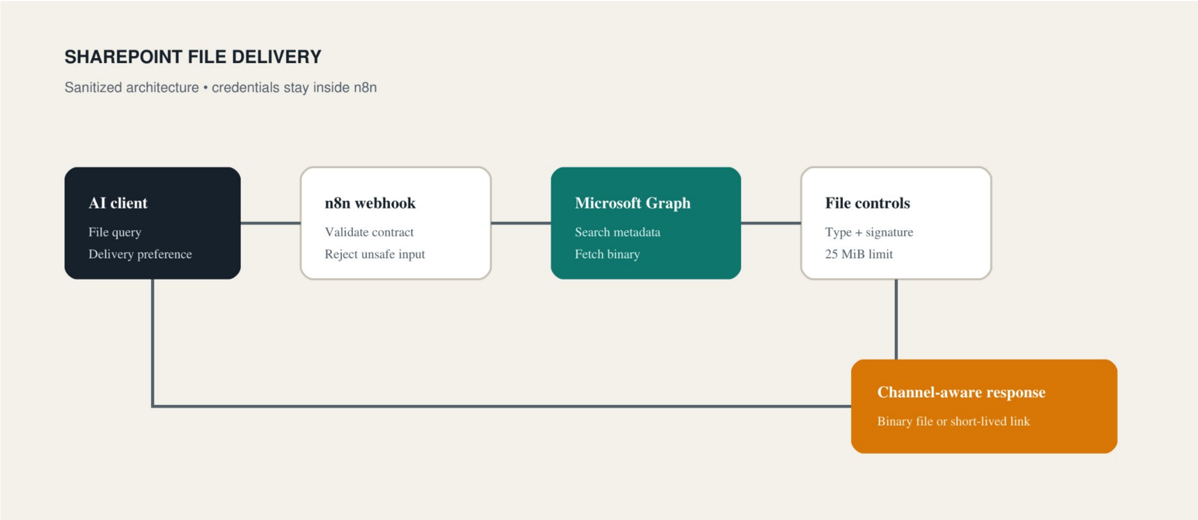
| System layer | Examples in this portfolio |
|---|---|
| Workflow and integration | n8n webhooks, Microsoft Graph, file delivery, deployment helpers |
| Research and evidence | due-diligence skills, source logs, evidence matrices, confidence and gap handling |
| Decision support | council seats, dissent registers, readiness gates, human decision rights |
| Documents and knowledge | DOCX generation, OOXML checks, Markdown conversion, retrieval indexes |
| Reporting and content | charts, recurring reports, transcript-to-storyboard pipelines, SEO operating systems |
| Product and media | TypeScript scoring logic, responsive website, voice parser, SVG infographic skill |

## Evidence strength

**Implemented / evaluated** means the private source project contains the working system and the portfolio publishes a safe extract or aggregate. **Runnable public proof** means a reviewer can execute the fixture locally. **Source extract** means authored code, captures, or generated artifacts are public while the complete deployment remains private. **Design-stage / scaffold** is included for completeness and is never promoted as finished.

The useful pattern across the work is consistent: define the contract, control the sources, make failure states explicit, preserve a human approval point, and leave an artifact that another person can inspect.

# Case 1 — SharePoint file delivery through n8n




SharePoint delivery architecture

## The operational problem

An AI assistant could answer questions but could not safely retrieve and return files held in SharePoint. The workflow needed to search by constrained input, authenticate without exposing credentials, handle Microsoft Graph responses, reject unsafe or oversized files, and adapt delivery to channels with different capabilities.

## What I implemented

The private n8n tool exposed separate webhook paths for file search and file retrieval. Code nodes normalized the caller request and required an exact safe match. Microsoft Graph requests ran through n8n-managed OAuth credentials. Before delivery, the workflow checked file metadata, actual binary size, MIME information, and extension consistency. A companion route created a temporary link when the calling channel could not accept a binary response.

The deployment layer updated workflows through the n8n API and supported a repeatable, idempotent correction path.

## Recorded proof

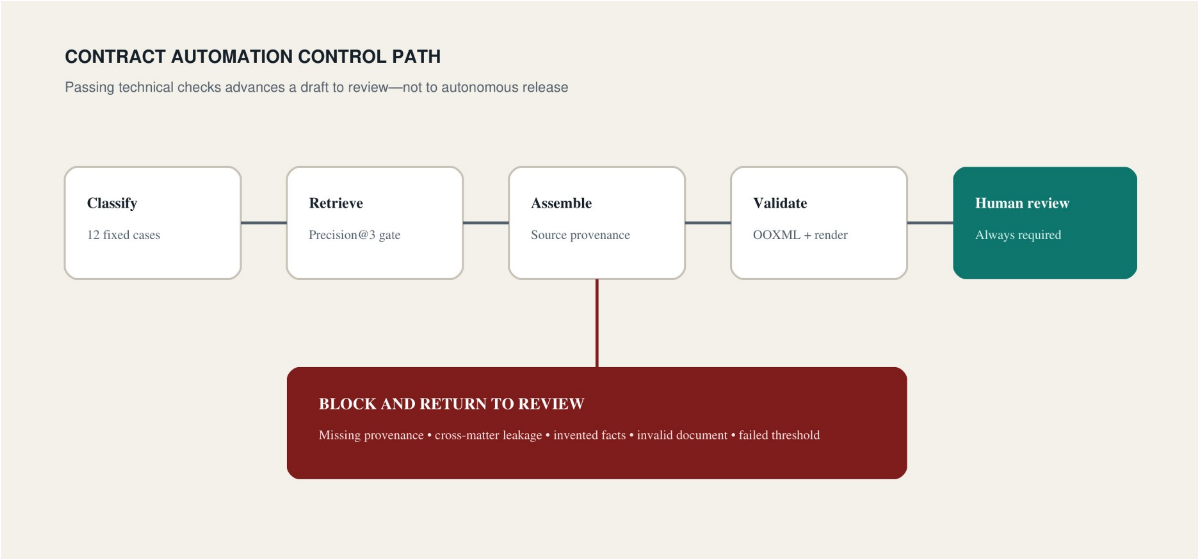
- `24,112,730`-byte audio file fetched successfully;
- `25 MiB` maximum response guard;
- byte-signature logic used to correct an extension mismatch;
- temporary PDF link flow exercised;
- idempotent workflow redeployment completed;
- three public unit tests pass.

## Public artifacts

`projects/n8n-deployment-and-tools/workflows/sharepoint-file-delivery.workflow.json` `projects/n8n-deployment-and-tools/src/deploy_workflow.py` `projects/n8n-deployment-and-tools/samples/` `projects/n8n-deployment-and-tools/tests/`

Production tenant IDs, workflow IDs, webhook paths, and credentials were replaced with fictional values.

# Case 2 — Governed contract automation




Contract controls

## The operational problem

Contract generation becomes risky when a model can invent material facts, pull language from the wrong matter, lose clause provenance, or produce a Word file that looks complete but fails structural checks. The system needed to accelerate assembly without confusing a technically valid draft with legally approved work.

## What I implemented

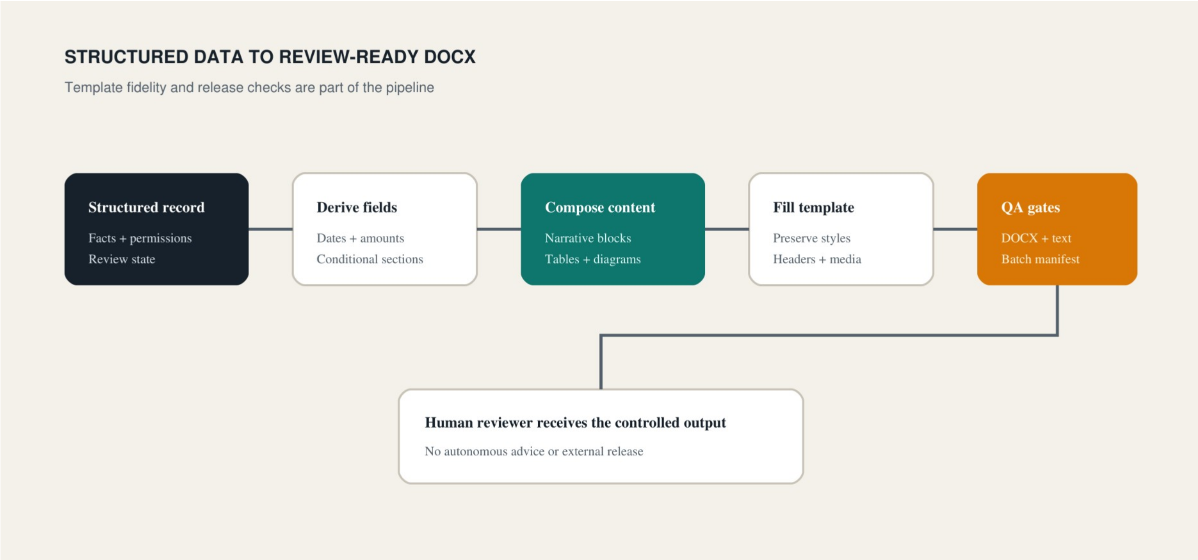
The private pipeline classifies the requested contract family, retrieves relevant clauses from a controlled precedent corpus, assembles a document package, records the source and selection reason, and leaves unknown material facts as review items. Negative tests deliberately remove required topics and annexures. Clean and review profiles apply different comment and marker rules. DOCX output is checked as an OOXML package and rendered for blank pages, density, and edge collisions.

## Fixed evaluation result

| Measure | Recorded result |
|---|---|
| Classification accuracy | 1.000 across 12 cases |
| Retrieval precision at 3 | 0.917 |
| Provenance coverage | 1.000 |
| Cross-matter leakage rate | 0.000 |
| Blind replay fact accuracy | 1.000 |
| Blind replay structural recall | 1.000 |
| Invented material facts | 0 |
| End-to-end rendered sample | 4 pages; deterministic checks passed |
| Operational reliance allowed | false |

The public package publishes aggregate results and a small runnable evaluation core, not precedent language or client agreements.

# Case 3 — Advice-document production pipeline




Advice-document pipeline

## The operational problem

Structured advice data had to become a readable document without losing the logic behind dates, amounts, allocations, permissions, risk language, conditional sections, or the integrity of the Word template.

## What I implemented

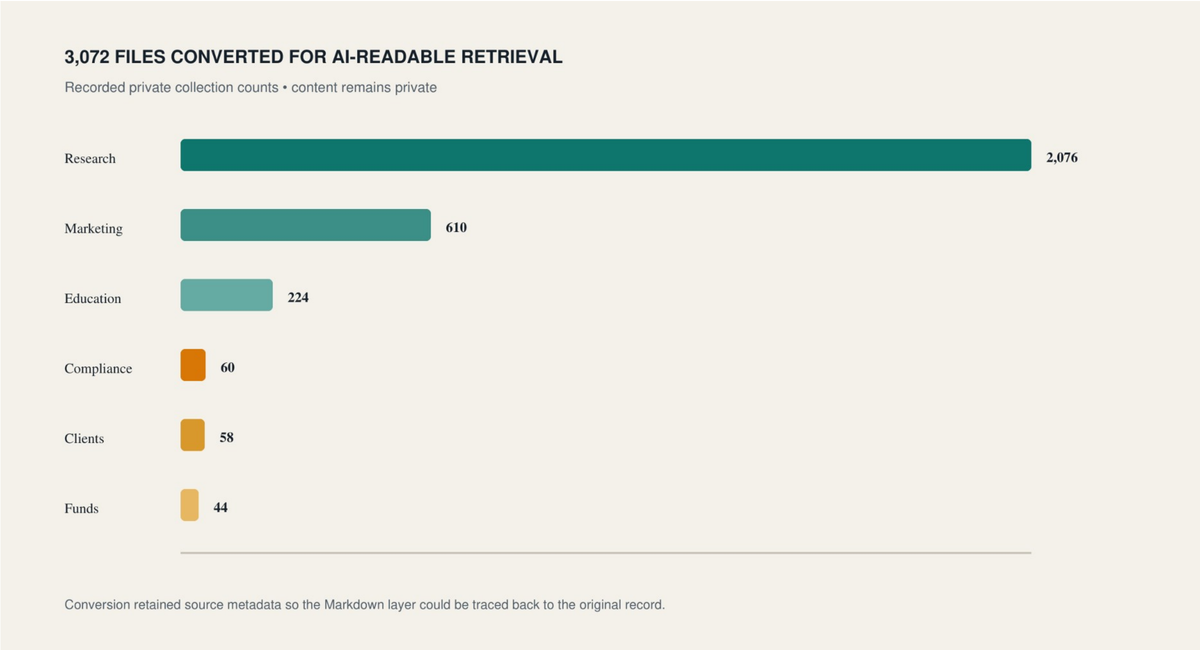
The generator derives fields from structured input, builds reusable narrative blocks, inserts markdown-style tables, suppresses non-applicable sections, removes internal codes, and fills a governed DOCX template. The template path preserves named styles, table shells, headers, footers, embedded images, and pagination controls. Batch QA checks manifests, generated Markdown, final DOCX files, unresolved placeholders, and expected output counts.

## Recorded proof

- active v8 generation path implemented;
- staged v9 batch: 12 work items and 24 DOCX files;
- recorded batch QA passed;
- v11 preserved styles, tables, headers, footers, and images;
- visible placeholder checks cleared;
- three public unit tests exercise the release blockers.

The public fixture uses a fictional person and allocation. It demonstrates composition and conditional logic only and is not financial advice.

# Knowledge and research systems




Knowledge collection

## AI-readable knowledge base

The conversion workspace transformed `.docx`, `.pdf`, `.txt`, `.md`, `.csv`, `.xlsx`, and `.pptx` sources into Markdown with frontmatter that retained file name, source path, URL, library, type, size, dates, authorship fields, record ID, hash, and conversion time. A master index grouped and described the resulting files.

The recorded private collection contained 3,072 files: 2,076 research, 610 marketing, 224 education, 60 compliance, 58 client, and 44 fund records. The portfolio includes the original-derived converters and a generated fictional mini corpus.

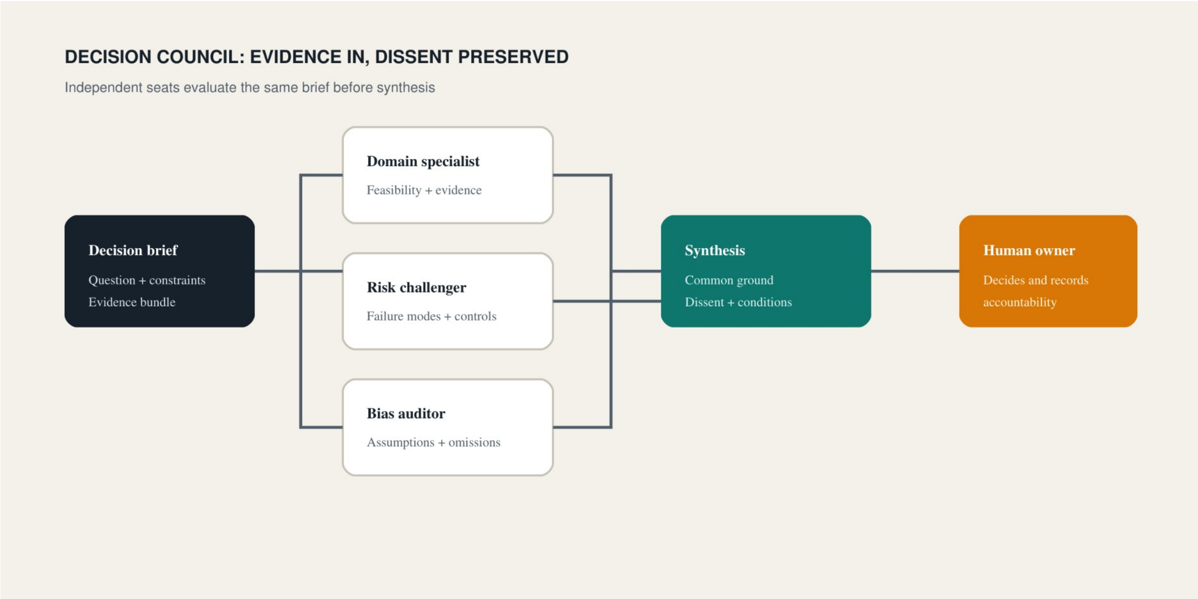
## Research skills and report builder

Separate agent routines covered company research, infrastructure readiness, and case-level due-diligence orchestration. Each routine defined current-source requirements, confidence, discrepancies, evidence gaps, and a structured human-review handoff.

The report-builder scripts create a research brief, source log, evidence matrix, and one of nine report archetypes before writing begins. The public demo stages a briefing-note workspace on AI document QA without making external research claims.

Together, these projects show a research approach that keeps collection, evidence, synthesis, and document design as distinct steps.

# Decisions and controlled handoffs




Decision council

## Multi-perspective council

A decision brief and evidence bundle activate independent specialist seats. Each seat must return a position, evidence used, risks, conditions, and confidence. Synthesis records common ground and preserves dissent; it does not silently average conflicting views into a generic recommendation.

The public validator rejects a single-seat output and requires a human decision gate. A fictional pilot decision demonstrates the full brief, two perspectives, disagreement, conditions, and unresolved ownership question.

## Assessment stage gates

The assessment state machine prevents a case from skipping readiness, research, evidence-gap handling, or human review. Missing identity, entity, ownership, mandate, or intended-activity evidence pauses the case. Code-level permission rules prevent automation from approving or declining.

## GitHub-to-n8n handoff

A filesystem contract separates the request, evidence log, and result. The validator confirms required files, matching case IDs, allowed result states, and a human-review flag before an automation consumes the package.

These three projects show the same core belief from different angles: orchestration should clarify accountability, not hide it.

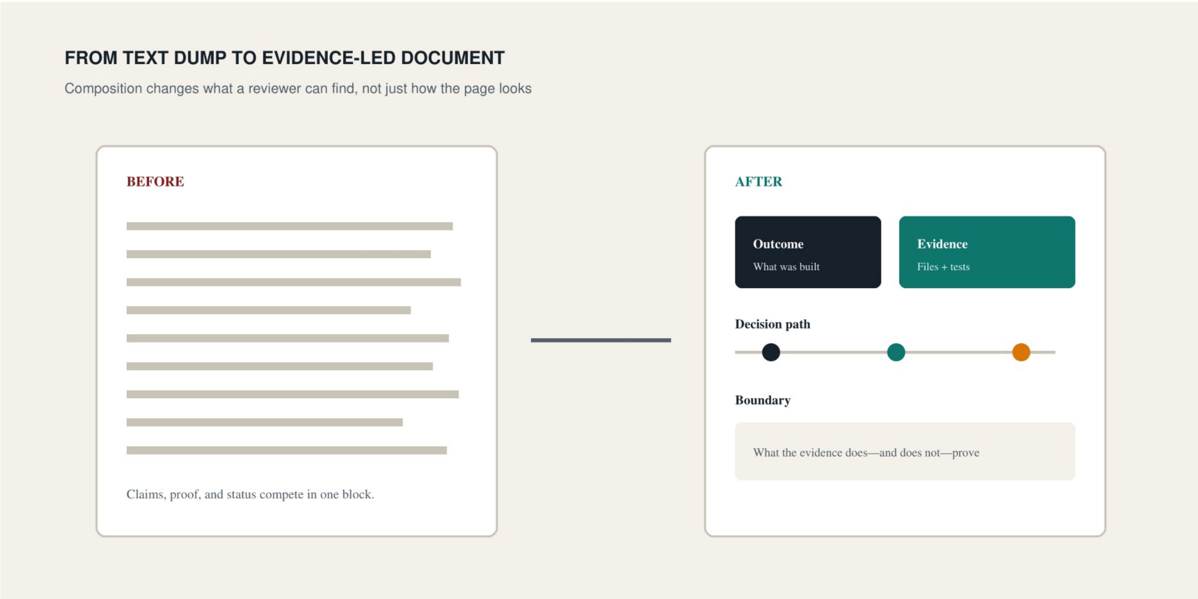
# Reports and document composition

## Market-report automation

The private workspace contains collectors for market commentary, token unlocks, and ETF-flow data, plus chart generators and Word report assembly. The public proof uses synthetic data and a dependency-free SVG renderer. It validates date order, requires at least two observations, labels the fixture, and includes a source manifest.

## Document composition system

The neutral public design system uses an ink/paper palette, teal for implemented paths, amber for approval or caution, and red for blocked states. Page archetypes distinguish recruiter scans, case studies, evidence pages, comparisons, and appendices. DOCX remains editable; PDF is the fixed review format; Markdown is canonical.




Composition before and after

## Annual reporting workspace

The annual-report project reuses the report builder, chart layer, and document QA rather than duplicating them. Its public integration manifest defines context, sources, data, charts, draft, output, and QA folders. The complete private report content remains excluded.

The result is a reporting method in which every chart has a dated source, every material statement can be mapped to evidence, and the final document is treated as a tested artifact.

# Content, learning, voice, and visuals

## Research to content

The pipeline separates evidence extraction, the writing brief, the script, and the storyboard. Timestamped transcript evidence becomes an approved claim; the script cannot silently add an unsupported idea; the storyboard maps each beat to a visual and on-screen message. The public fictional fixture and two tests demonstrate traceability and missing-topic failure.

## Prompt modules

A large education workflow was decomposed into shared source-bound policy plus quiz, flashcard, and FAQ modules. Each module has one output contract. Workflow IDs and course material were removed. Fictional outputs show how the same small source becomes distinct learning artifacts.

## Voice and podcast renderer

The original parser accepts a two-speaker line contract, chunks long turns, and keeps speaker assignment stable. The audio layer converts to mono, inserts pauses, checks sample-rate consistency, normalizes peaks, and exports PCM WAV. Voice references and model weights are excluded from the public package.

## SVG infographic skill

The complete authored skill includes semantic source syntax, rendering instructions, editable SVG output, and a WEBP preview. It demonstrates that a reusable AI skill can ship with both instructions and a tangible artifact.

The LearnWorlds workspace is separately labeled as a scaffold because the scanned source folder did not contain the expected publishing code.

# Product, website, and SEO work

## Club 4301

The running-club MVP recognizes contribution beyond pace. Original TypeScript code defines six scoring categories, applies awarded-task points, calculates totals, and builds recap data. The retained source Vitest run passed. The complete application and deployment environment remain private pending a dedicated release review.

## AI education site

The site organizes beginner AI learning into five pillars: fundamentals, practical skills, workflows, career readiness, and responsible use. Final local desktop and mobile captures show the responsive layout and content hierarchy.

The accompanying audit converted site issues into specific actions: repair a draft destination from cornerstone content, add incoming links to 11 orphaned article records, publish missing hub topics before dependent pages, differentiate overlapping keyword pairs, remove production notes from public copy, and verify indexing state in Search Console. No ranking or traffic outcome is claimed.

## SEO automation architecture

The design-stage workflow defines keyword intake, intent, brief, sources, draft, quality, internal links, metadata, human approval, and publication handoff. A promotion checklist states what is still missing: a sanitized runnable workflow, cached fixtures, negative tests, and a non-production publication receipt.

This section demonstrates both delivery and restraint: screenshots and audits are proof; a diagram alone is not called implementation.

# How I work

## 1. Define the handoff

Inputs, allowed states, outputs, and owners are written before the automation is treated as complete.

## 2. Bind claims to evidence

Research findings retain sources and dates. Document systems preserve provenance. Fictional fixtures are labeled. Missing evidence stays visible.

## 3. Test the failure path

The public tests reject insecure URLs, cross-matter leakage, unresolved placeholders, invalid state transitions, incomplete handoffs, unsorted data, missing topics, and unsupported speakers.

## 4. Keep the human decision

Contract reliance, applicant decisions, advice, publication, and external release remain human responsibilities. The automation prepares, validates, and records the work.

## 5. Publish inspectable proof

The repository includes source, workflow JSON, schemas, fictional inputs and outputs, diagrams, screenshots, test receipts, Markdown, DOCX, and PDF. Private staging and hashes remain ignored.

## Contact and next step

GitHub: `https://github.com/ChrisBuen`

For a technical review, start with the three flagship folders and run their standard-library tests. For a recruiter scan, use the two-page brief. For full breadth, use the evidence appendix and project catalog.